

UNZENDAKE, Japan

WMO: 478180

Lat: 32.7378N

Lon: 130.2625E

Elev: 2230

StdP: 13.55

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.7	26.2	14.2	12.3	31.6	16.8	13.9	31.2	26.8	41.6	23.1	36.4	9.4	300	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	83.3	73.7	81.6	73.2	79.7	72.5	75.6	79.9	74.7	78.5	73.8	77.0	6.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	140.6	77.4	73.7	136.3	76.1	72.9	132.5	75.3	41.0	80.5	40.0	78.9	39.1	77.3	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	23.9	20.4	20.1	86.6	2.6	1.9	18.2	88.0	16.7	89.1	15.2	90.1	13.3	91.4
			18.6	78.0	2.6	1.3	16.7	78.9	15.2	79.6	13.7	80.3	11.8	81.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	36.4	38.6	44.1	52.8	60.8	66.7	73.2	74.8	69.3	60.3	50.2	40.4	(6)
		DBStd	14.17	5.80	6.58	6.16	5.65	3.95	3.28	3.01	2.77	4.27	5.11	6.33	6.32	(7)
		HDD50	1363	424	322	200	36	0	0	0	0	2	76	303		(8)
		HDD65	4175	888	738	647	366	137	19	0	12	161	445	762		(9)
		CDD50	3456	1	4	18	120	336	502	720	768	579	321	81	6	(10)
		CDD65	794	0	0	0	0	8	72	255	303	141	15	0	0	(11)
		CDH74	2714	0	0	0	0	23	84	883	1342	365	17	0	0	(12)
		CDH80	336	0	0	0	0	1	2	117	195	21	0	0	0	(13)
(14)	Wind	WSAvg	8.0	8.0	8.3	8.7	8.5	7.8	7.5	8.4	7.6	7.7	7.8	7.6	8.4	(14)
(15)	Precipitation	PrecAvg	100.6	3.1	4.5	7.2	8.2	9.2	20.1	17.2	9.7	9.1	4.6	4.3	3.5	(15)
(16)		PrecMax	166.3	7.0	9.9	16.4	14.2	18.5	39.7	39.3	27.0	24.7	11.3	9.5	6.5	(16)
(17)		PrecMin	55.0	0.5	2.0	3.7	3.5	2.4	4.8	2.3	2.1	2.0	0.4	0.8	0.1	(17)
(18)		PrecStd	21.3	1.5	1.9	2.7	3.1	4.1	8.8	9.3	5.9	5.8	3.0	2.3	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.8	59.3	62.9	70.8	76.9	78.7	85.2	86.1	82.4	76.4	67.1	58.9		(19)
(20)		0.4% MCWB	54.3	54.9	53.6	57.5	60.5	67.0	73.9	74.6	71.8	66.6	60.3	54.0		(20)
(21)		2% DB	51.8	56.0	59.6	66.9	73.5	76.4	83.0	83.8	80.1	72.8	64.5	56.0		(21)
(22)		2% MCWB	49.2	53.1	52.8	56.4	60.0	66.2	73.8	74.0	71.0	65.3	59.6	52.0		(22)
(23)		5% DB	48.9	53.1	56.9	64.3	71.0	74.4	81.1	82.3	78.0	70.6	62.3	53.6		(23)
(24)		5% MCWB	45.9	50.2	52.0	56.9	59.9	66.7	73.4	73.7	70.4	64.0	58.4	50.1		(24)
(25)		10% DB	45.8	49.7	54.4	62.0	68.9	72.6	78.9	80.5	75.9	68.5	60.1	50.9		(25)
(26)		10% MCWB	42.3	46.2	50.4	55.3	59.9	67.7	72.8	73.2	69.4	62.7	56.2	47.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	54.6	57.8	59.0	64.6	68.6	72.9	77.3	77.4	74.8	70.9	64.1	57.2		(27)
(28)		0.4% MCDB	55.4	57.9	59.3	65.6	69.7	73.5	81.5	81.9	78.3	72.9	64.9	57.4		(28)
(29)		2% WB	50.4	54.0	55.9	62.0	66.6	72.0	75.6	76.0	73.3	68.7	61.6	53.6		(29)
(30)		2% MCDB	51.2	54.9	57.2	63.7	68.1	72.7	79.7	80.6	76.4	71.1	62.9	54.8		(30)
(31)		5% WB	46.5	50.7	53.5	59.6	64.7	71.1	74.5	75.0	72.3	66.6	59.2	50.8		(31)
(32)		5% MCDB	48.1	52.6	55.3	61.4	66.8	72.0	78.1	79.3	75.5	69.0	61.4	52.8		(32)
(33)		10% WB	42.8	47.1	51.0	57.4	62.9	70.0	73.7	74.2	71.3	64.4	56.9	47.8		(33)
(34)		10% MCDB	45.2	49.0	53.7	60.3	66.0	71.2	76.5	78.1	74.5	67.1	59.6	50.4		(34)
(35)	Mean Daily Temperature Range	MDBR	10.9	12.4	13.5	14.2	14.3	10.4	9.2	10.4	11.5	13.0	12.4	11.5		(35)
(36)		5% DB MCDBR	14.5	15.3	16.2	16.7	18.4	14.5	13.7	13.6	13.7	14.7	12.8	14.1		(36)
(37)		5% DB MCWBR	12.9	13.7	12.4	10.9	10.7	8.1	7.3	6.8	7.3	8.8	9.1	11.5		(37)
(38)		5% WB MCDBR	13.2	13.6	13.3	11.4	11.6	6.8	11.2	10.6	10.6	10.2	10.6	12.3		(38)
(39)		5% WB MCWBR	13.3	13.8	12.8	10.3	9.8	5.6	6.6	6.3	6.2	7.7	9.4	11.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.447	0.503	0.521	0.530	0.539	0.530	0.507	0.477	0.432	0.404	0.400		(40)
(41)		taud	2.078	2.011	1.869	1.876	1.915	1.977	2.052	2.114	2.147	2.182	2.180	2.117		(41)
(42)		Ebn at Noon	246	250	246	248	247	244	245	249	251	253	247	242		(42)
(43)		Edh at Noon	42	49	61	63	62	58	53	49	46	41	37	38		(43)
(44)	All-Sky Solar Radiation	RadAvg	719	947	1281	1528	1629	1352	1588	1658	1367	1148	845	669		(44)
(45)		RadStd	75	88	83	134	183	209	216	152	144	118	83	51		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+1.31	N/A	+0.83	+1.04	N/A	N/A	N/A	N/A	(46)
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page